[METRIC]
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SUPERSEDING
MIL-N-62418(AT)
6 December 1992

COMMERCIAL ITEM DESCRIPTION

NICKEL BASE ALLOY, SHEET AND PLATE CORROSION AND HEAT RESISTANT, SOLUTION HEAT TREATED (HAYNES 230) (METRIC)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers the chemical composition and mechanical properties of nickel base alloy in form of sheet and plate. This material is used primarily for formed and drawn parts requiring high strength up to 980 degrees Celsius (°C) (1800 degrees Fahrenheit (°F)), and oxidation resistance up to 1095°C (2000°F).

2. CLASSIFICATION. The nickel base alloy covered in this specification shall be classified as follows (see 3.1 and 7.2):

Type Ia - Hot rolled sheet
Type Ib - Cold rolled sheet
Type II - Hot rolled plate

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 9630

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3. SALIENT CHARACTERISTICS

3.1 Materials. The nickel base alloy shall come in three types (see 2) and shall be solution heat treated (see 4.1).

3.1.1 Finish. The surface of the sheet or plate shall be delivered descaled unless solutionizing produces a bright finish of equal or better quality than the no. 1 finish for corrosion resistant sheet (see 7.2).

3.1.2 Chemical composition. The alloy shall be nickel base, unified numbering system (UNS) N06230 in the forms of sheet and plate. The chemical composition of the alloy shall conform to the percentage by weight of each element as specified in ASTM B435 or equivalent (see 7.2).

3.1.2.1 Chemical composition variation. Allowable variation of the chemical composition shall be in accordance with the requirements of ASTM B435.

3.1.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.1).

3.2 Tolerances. Unless otherwise specified in the contract or purchase order (see 7.2) tolerances shall be in accordance with the requirements of AMS 2262.

3.3 Performance. The material shall conform to the following mechanical and physical properties after heat treatment.

3.3.1 Mechanical properties. Mechanical properties of the heat treated material shall be as specified herein.

3.3.1.1 Tensile. The room temperature tensile properties of the nickel base alloy shall be as follows and in accordance with ASTM B435:

	Megapascals	(psi)
Ultimate strength, minimum:	758	110 000
Yield strength at 0.2% offset, minimum:	310	45 000
Elongation in 50.8 millimeter (mm) 2 inch (in.), minimum thickness 0.38 mm (0.015 in.) and over	Elongation	
	40%	40%

NOTE: For material thicknesses less than 0.38 mm (0.015 in.), elongation shall be as agreed upon by the procuring activity and the supplier.

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3.3.1.2 Stress rupture. A test specimen maintained at a temperature of 927 ±2°C (1700 ± 3°F), while the axial load specified below is applied continuously, shall not rupture in less than 24 hours. Elongation after rupture, measured at room temperature, shall be not less than that specified below.

Nominal thickness mm (in.)	Stress kPa (psi)	Elongation in 2 in. %, min.
Up to 0.50 (0.020)	48 (7.000)	5
Over 0.50 (0.020)	61 (9.000)	10

3.3.1.3 Bending. All bending shall be done at room temperature and in accordance with ASTM E290, by aligning the rolling direction parallel to the bending axis, and wrapping the material through an angle of 3.15 radians (180 degrees). To calculate bend radius multiply the nominal material thickness by the appropriate bend factor listed below. No cracking is permitted.

Nominal thickness	Bend factor
Up to 1.25 mm (0.050 in.)	1.5
1.25 to 4.75 mm (0.050 to 0.187 in.)	2.0
4.75 mm (0.187 in.) and above	As specified in contract or purchase order (see 7.2)

3.3.2 Physical properties. Physical properties of the heat treated material shall be as specified herein.

3.3.2.1 Grain size. The grain size of the nickel base alloy shall be sectioned and polished to appropriate finess and suitably etched to reveal grain structure. The grain size shall be of the following sizes and in accordance with ASTM E112:

Nominal thickness	ASTM grain size
Up to 3.2 mm (0.125 in.)	4 or finer
Over 3.2 mm (0.125 in.)	3 or finer

3.4 Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), the heat number, the alloy nominal thickness, the UNS number N06230, and the part identification number (PIN) (see 7.2 and 7.4).

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4. REGULATORY REQUIREMENTS

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, including examinations and tests.

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract or order.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Non-Government publications. Copies of AMS 2262, "Tolerances Nickel, Nickel Alloy, and Cobalt Alloy Sheet, Strip, and Plate (DoD Adopted)" are available from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001. Copies of ASTM E112, "Standard Test Methods for Determining Average Grain Size (DoD Adopted)"; ASTM E290, "Standard Test Methods for Semi-Guided Bend Test for Ductility of Metallic Materials (DoD Adopted)"; and ASTM B435, "Standard Specification for UNS N06002, UNS N06230, UNS N12160, and UNS R30556 Plate, Sheet, and Strip are available from American Society For Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of DODISS to be cited in the CID, and if required, the specific issue of individual documents referenced.

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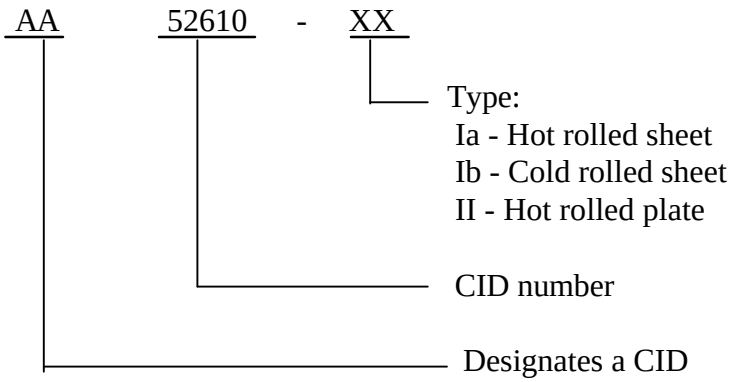
c. Chemical composition, if applicable, for oversize.
d. Type of nickel base alloy plate or sheet.
e. If finish other than as specified.
f. If tolerances shall be other than as specified.
g. Bend factor for material greater than 4.75 mm.
h. Identification and marking.
i. Selection of applicable level and packaging requirements.

7.3 Cross-reference data. Alloy conforming to this CID is interchangeable/substitutable with alloy conforming to MIL-N-62418(AT) and Textron Lycoming internal specification M3633B, dated 13 June 1991.

7.4 Part identification number (PIN). The PINs to be used for nickel base alloy required to this CID are created as follows:



MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 9630-0039)

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